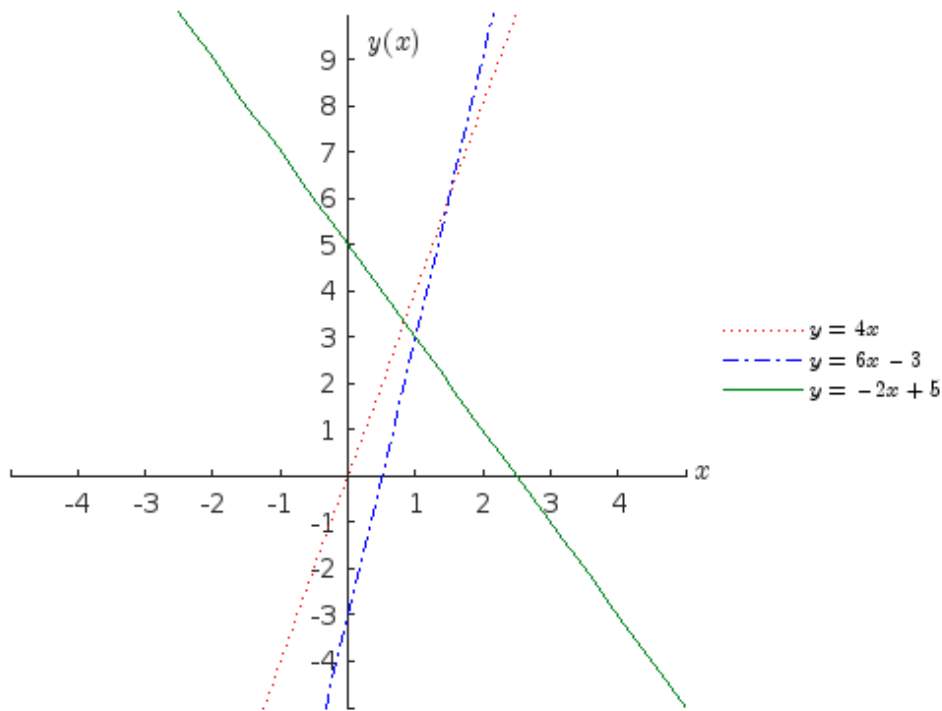


3.2 Linear Functions

Linear functions are among the simplest and most important functions encountered in mathematics, engineering and science. Many physical systems may be approximated by linear relationships over a limited range. Examples include:

- Distance travelled at constant speed.
- Cost as a function of quantity purchased.
- Electrical relationships such as Ohm's Law.
- Conversion between temperature scales.

The graph of a linear function is always a straight line, for example



3.2.1 The General Form of a Linear Function

Definition

A linear function may be written in the form

$$y = mx + c$$

where m is the gradient and c is the y -intercept.

The variable x is the independent variable and y is the dependent variable.

3.2.2 The Gradient

The gradient describes the steepness of a line.

Definition

The gradient of a straight line is

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

where Δy is the vertical change and Δx is the horizontal change.

The gradient tells us how much y changes when x increases by one unit.

Worked Example

Determine the gradient of the line passing through the points $(0, 2)$ and $(6, 14)$.

Here $(x_1, y_1) = (0, 2)$ and $(x_2, y_2) = (6, 14)$.

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{14 - 2}{6 - 0} \\ &= \frac{12}{6} \\ &= 2 \end{aligned}$$

3.2.3 Positive and Negative Gradients

A line with positive gradient ($m > 0$) rises from left to right. A line with negative gradient ($m < 0$) falls from left to right. If the graph is horizontal, the gradient is zero ($m = 0$).

Worked Example

Determine the gradient of the line passing through $(2, 10)$ and $(10, -6)$.

Here $(x_1, y_1) = (2, 10)$ and $(x_2, y_2) = (10, -6)$.

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{-6 - 10}{10 - 2} \\ &= \frac{-16}{8} \\ &= -2 \end{aligned}$$

Therefore the gradient is $m = -2$. The line decreases from left to right.

3.2.4 The y -Intercept

The y -intercept is the value of y when $x = 0$.

Definition

For the linear function

$$y = mx + c$$

the constant c is the y -intercept.

The graph crosses the y -axis at

$$(0, c)$$

Worked Example

Determine the y -intercept of $y = 3x + 4$.

Comparing with $y = mx + c$ gives $c = 4$. Therefore the line crosses the y -axis at $(0, 4)$.

Worked Example

Determine the y -intercept of $y = -5x + 7$.

Alternatively, the line crosses the y -axis at $x = 0$. Substituting gives:

$$y = mx + c$$

$$y = -5x + 7$$

$$y = -5 \times 0 + 7$$

$$y = 7$$

Therefore the co-ordinates of the y -intercept are $(0, 7)$.

3.2.5 The x -Intercept

The x -intercept occurs where the graph crosses the x -axis. At this point $y = 0$.

Worked Example

Find the x -intercept of $y = 2x + 8$.

Let $y = 0$.

$$0 = 2x + 8$$

$$-8 = 2x$$

$$x = -4$$

Therefore the graph crosses the x -axis at $(-4, 0)$.

3.2.6 Sketching Linear Functions

To sketch a linear function:

1. Determine the y -intercept.
2. Find a second point (preferably the x -intercept).
3. Draw a straight line through the points.

Worked Example

Sketch $y = 2x + 1$.

When $x = 0$,

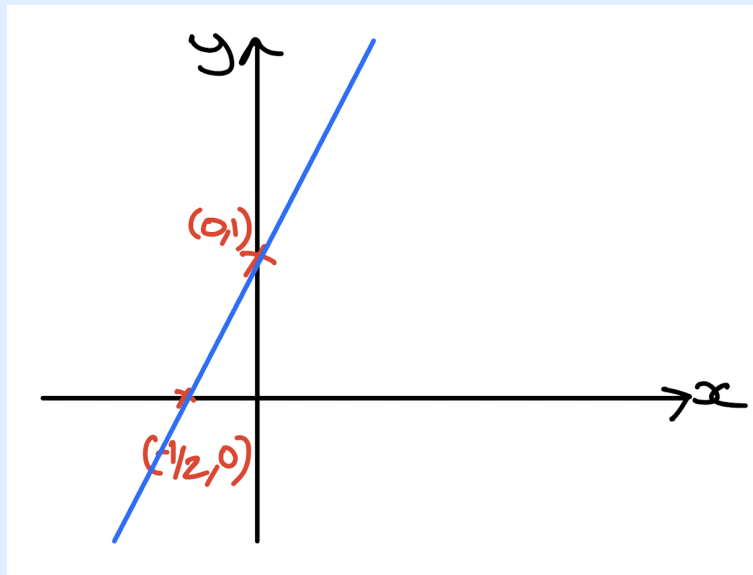
$$\begin{aligned}y &= 2(0) + 1 \\&= 1\end{aligned}$$

giving the y -intercept $(0, 1)$.

When $y = 0$

$$\begin{aligned}0 &= 2x + 1 \\-1 &= 2x \\x &= -\frac{1}{2}\end{aligned}$$

giving the point (x -intercept) $(-\frac{1}{2}, 0)$. Plotting $(0, 1)$ and $(-\frac{1}{2}, 0)$ and drawing a straight line through them produces the following graph:



3.2.7 Finding the Equation of a Straight Line

Sometimes the graph of a line is known and we wish to determine its equation. The procedure is:

1. Determine the gradient.
2. Determine the y -intercept.
3. Substitute into the general form of a linear line: $y = mx + c$.

Worked Example

A line passes through $(1, 3)$ and $(5, 11)$. Find its equation.

First determine the gradient.

$$\begin{aligned} m &= \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{11 - 3}{5 - 1} \\ &= 2 \end{aligned}$$

Substitute into $y = mx + c$ to give

$$y = 2x + c$$

Use the point $(1, 3)$ to find c .

$$\begin{aligned} y &= mx + c \\ 3 &= 2(1) + c \\ 3 &= 2 + c \\ c &= 1 \end{aligned}$$

Therefore $y = 2x + 1$.

3.2.8 Parallel Lines

Parallel lines have the same gradient.

Definition

Two straight lines are parallel if they have equal gradients.

For example,

$$y = 2x + 3$$

and

$$y = 2x - 5$$

are parallel because both have gradient $m = 2$.

3.2.9 Engineering Application

Engineering Application

For constant speed motion,

$$d = vt$$

where d is distance, v is speed and t is time. Suppose a train travels at 30 m/s. Then

$$d = 30t$$

3.2.10 Common Mistakes

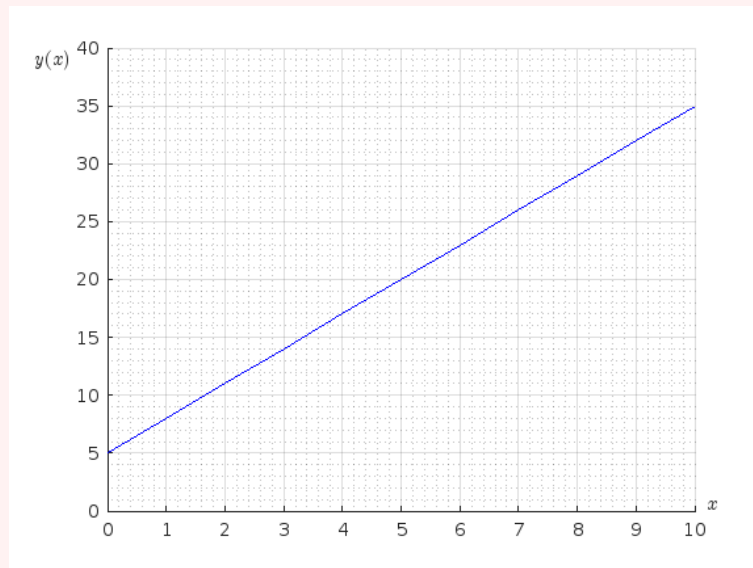
Common errors include:

- Confusing the gradient and intercept.
- Forgetting that the gradient is $\frac{\Delta y}{\Delta x}$ and not $\frac{\Delta x}{\Delta y}$.
- Using only one coordinate when calculating a gradient.
- Forgetting to set $y = 0$ when finding an x -intercept.
- Assuming all straight lines pass through the origin.

3.2.11 Exercises

Exercise

1. State the gradient and y -intercept of:
 - (a) $y = 3x + 2$
 - (b) $y = -4x + 1$
 - (c) $y = 7 - 2x$
2. Determine the gradient of the line passing through:
 - (a) $(1, 2)$ and $(4, 8)$
 - (b) $(2, 5)$ and $(6, -3)$
3. Find the equation of the line with:
 - (a) gradient 3 and intercept 2
 - (b) gradient -2 and intercept 5
4. Determine the equation of the line from the following graph:



5. Find the x -intercept of:
 - (a) $y = 2x + 8$
 - (b) $y = 5 - x$

Exercise

6. Sketch:

(a) $y = x + 2$

(b) $y = -2x + 4$

7. Determine whether the following pairs of lines are parallel:

(a) $y = 3x + 1$ and $y = 3x - 4$

(b) $y = 2x + 3$ and $y = -2x + 3$

8. A train travels at a constant speed of 25 m/s.

(a) Write a function relating distance d and time t .

(b) Determine the distance travelled after:

i. 20 s

ii. 40 s

iii. 120 s

3.2 Solutions to Exercise 3.2.11

1. A linear function has the form $y = mx + c$ where m is the gradient and c is the y -intercept.

(a) $y = 3x + 2$. Therefore,

$$m = 3, \quad c = 2$$

(b) $y = -4x + 1$. Therefore,

$$m = -4, \quad c = 1$$

(c) Rewrite in the form $y = mx + c$: $y = -2x + 7$. Therefore,

$$m = -2, \quad c = 7$$

2. Using

$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

(a) $(1, 2)$ and $(4, 8)$

$$\begin{aligned} m &= \frac{8 - 2}{4 - 1} \\ &= \frac{6}{3} \\ &= 2 \end{aligned}$$

Therefore, $m = 2$.

(b) $(2, 5)$ and $(6, -3)$

$$\begin{aligned} m &= \frac{-3 - 5}{6 - 2} \\ &= \frac{-8}{4} \\ &= -2 \end{aligned}$$

Therefore, $m = -2$.

3. Using $y = mx + c$

(a) Given $m = 3$ and $c = 2$

$$\begin{aligned} y &= mx + c \\ &= 3x + 2 \end{aligned}$$

Therefore, $y = 3x + 2$.

(b) Given $m = -2$ and $c = 5$

$$\begin{aligned} y &= mx + c \\ &= -2x + 5 \end{aligned}$$

Therefore, $y = -2x + 5$.

4. From the graph:

- The y -intercept is 5.

- The gradient is can be calculated by taking any two points: $(0, 5)$ and $(10, 35)$.
Therefore $m = \frac{35-5}{10-0} = 3$.

Therefore, $y = 3x + 5$.

5. The x -intercept occurs when $y = 0$

(a)

$$0 = 2x + 8$$

$$-8 = 2x$$

$$x = -4$$

Therefore, the x -intercept is $(-4, 0)$.

(b)

$$0 = 5 - x$$

$$x = 5$$

Therefore, the x -intercept is $(5, 0)$.

6. (a) For $y = x + 2$ the y -intercept= 2. Choose two points:

$$x = 0 \quad y = 2$$

$$x = 2 \quad y = 2 + 2 = 4$$

The graph is a straight line through $(0, 2)$ and $(2, 4)$. It is generally useful to also calculate the x -intercept. Thus,

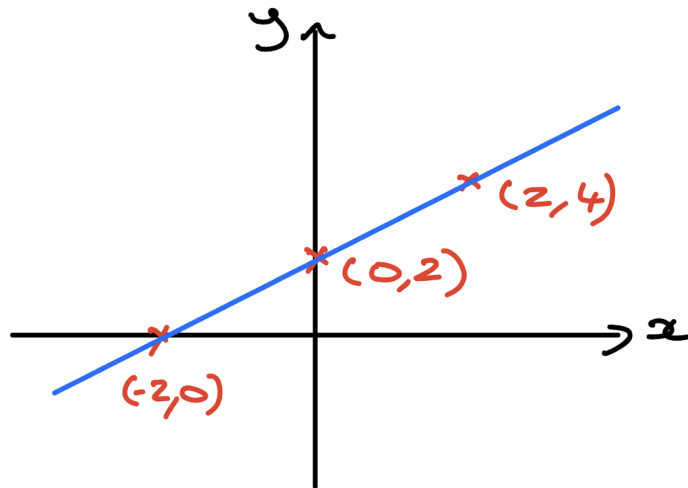
$$y = x + 2$$

$$\text{let } y = 0$$

$$0 = x + 2$$

$$x = -2$$

The graph looks as follows:



(b) For $y = -2x + 4$ the y -intercept = 4. Choose two points:

$$x = 0 \quad y = 4$$

$$x = 3 \quad y = -2 \times 3 + 4 = -2$$

The graph is a straight line through $(0, 4)$ and $(3, -2)$. Calculating the x -intercept yields

$$y = -2x + 4$$

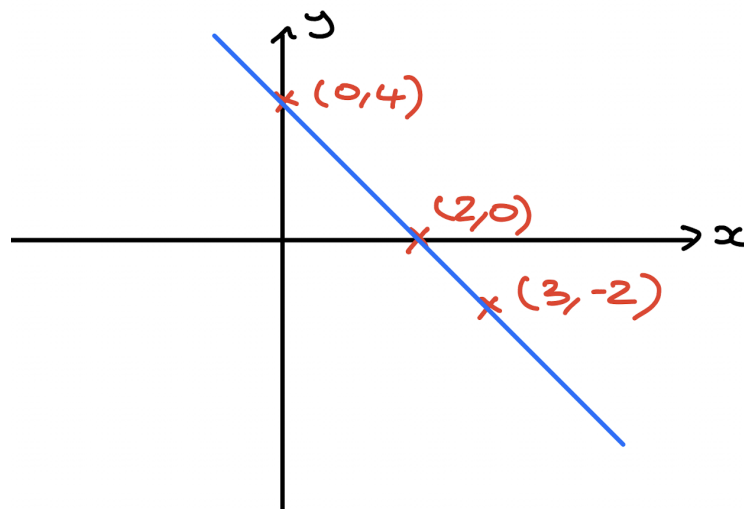
$$\text{let } y = 0$$

$$0 = -2x + 4$$

$$2x = 4$$

$$x = 2$$

The graph looks as follows:



7. Parallel lines have equal gradients.

(a) $y = 3x + 1$ has gradient $m = 3$ and $y = 3x - 4$ has gradient $m = 3$. Since the gradients are equal, the lines are parallel.

(b) $y = 2x + 3$ has gradient $m = 2$ and $y = -2x + 3$ has gradient $m = -2$. Since the gradients are different, the lines are not parallel.

8. (a) Distance travelled is given by

$$d = vt$$

Substituting $v = 25 \text{ m/s}$ gives $d = 25t$

(b) i.

$$\begin{aligned} d &= 25(20) \\ &= 500 \end{aligned}$$

Therefore, $d = 500 \text{ m}$.

ii.

$$\begin{aligned} d &= 25(40) \\ &= 1000 \end{aligned}$$

Therefore, $d = 1000 \text{ m}$.

iii.

$$\begin{aligned} d &= 25(120) \\ &= 3000 \end{aligned}$$

Therefore, $d = 3000 \text{ m}$.